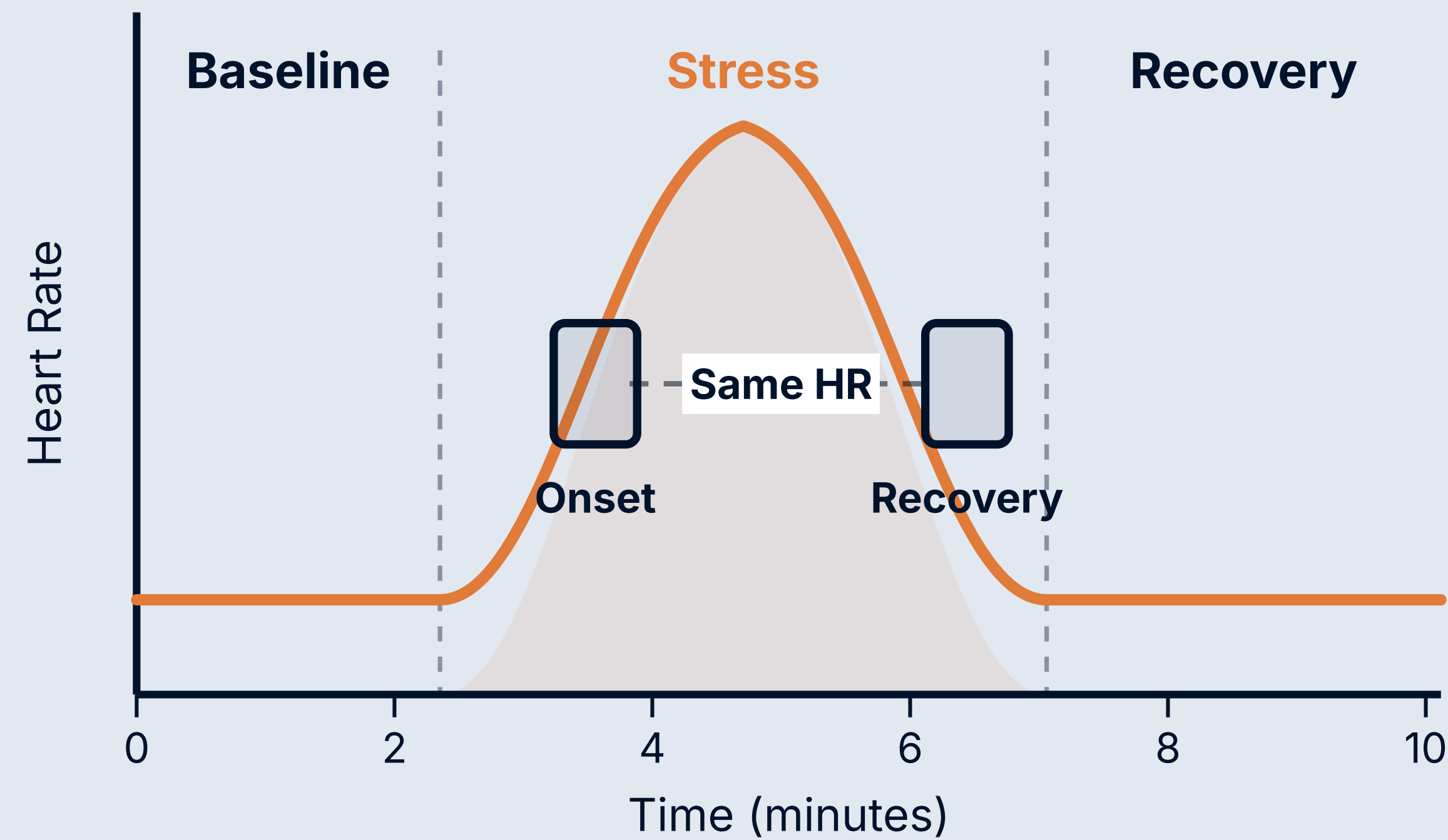




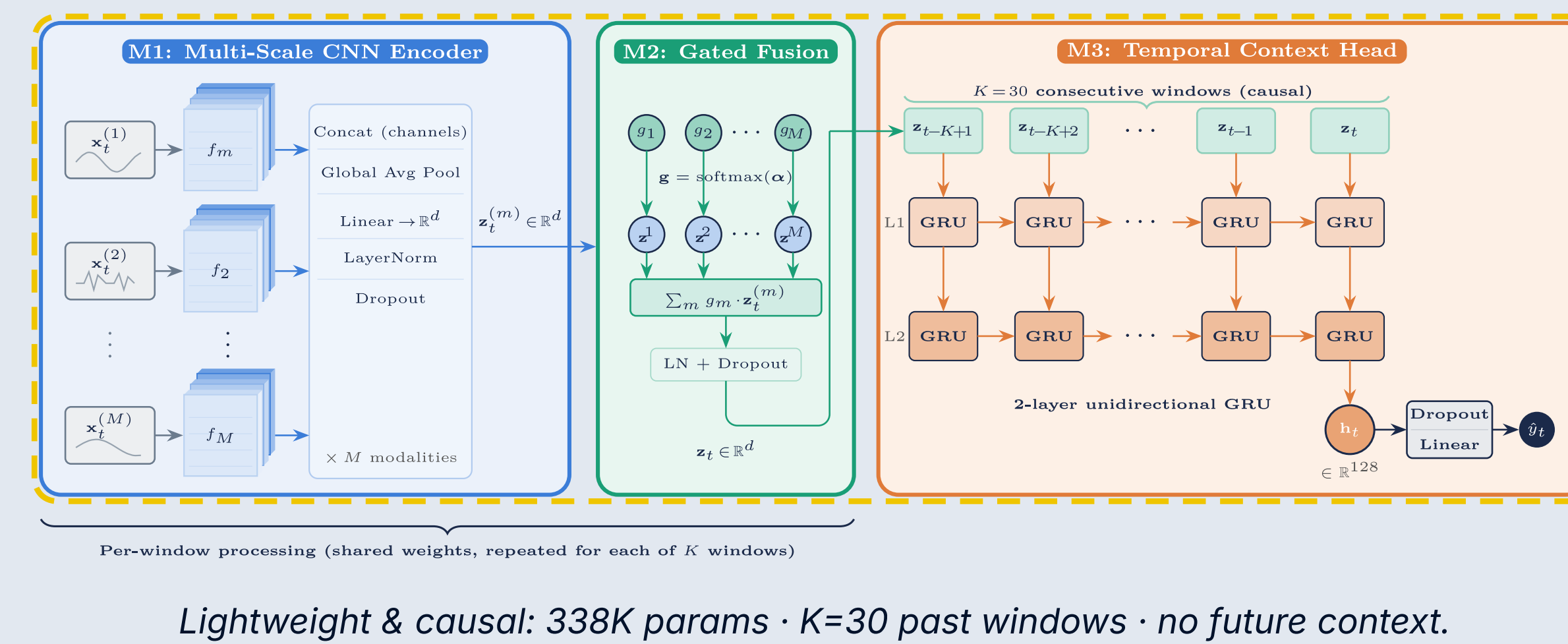
MOTIVATION

- **Physiological patterns span multiple time scales.** Heart rate changes rapidly; EDA and temperature drift slowly. Single-kernel CNNs capture one scale, not both → **M1**: multi-scale CNN encoder.
- **Sensors vary in informativeness.** EDA tracks arousal; ECG is sensitive to motion artifacts; sampling rates vary by 175× across sensors. Equal-weight fusion ignores this → **M2**: learned modality gates.
- **Stress evolves over time.** Stress develops gradually and persists, but window-level models classify 1-second snapshots in isolation → **M3**: causal temporal context over $K=30$ past windows.



A model sees the onset and recovery windows as identical (same HR) — only temporal context (M3) disambiguates them.

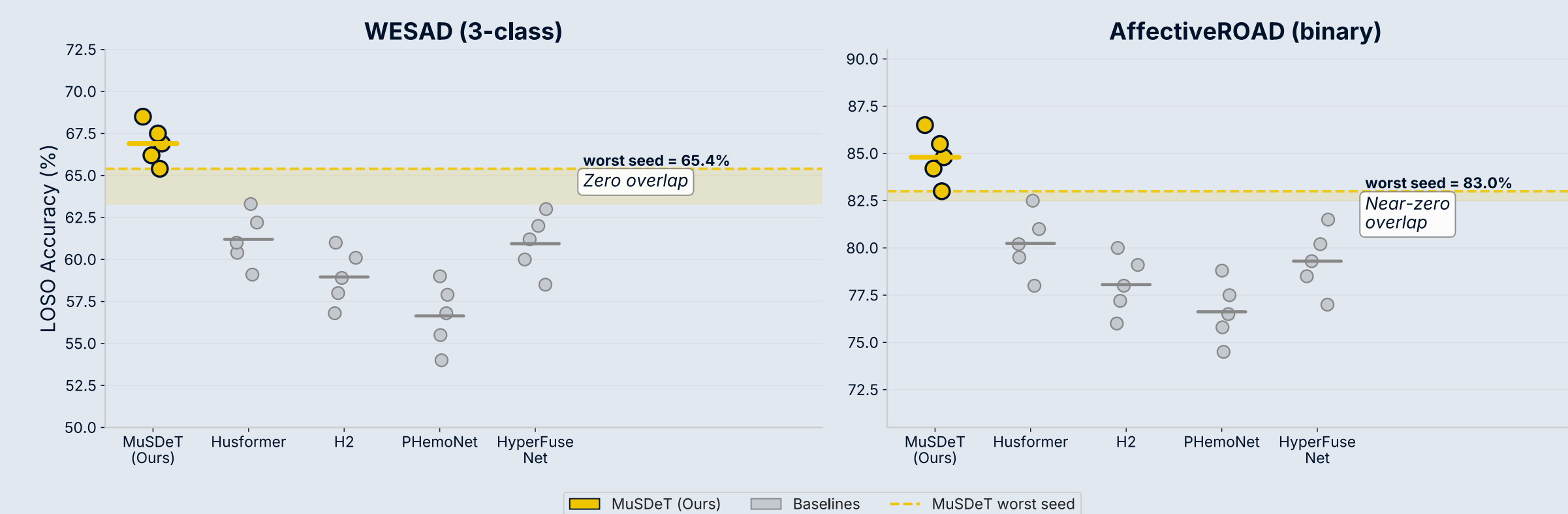
ARCHITECTURE



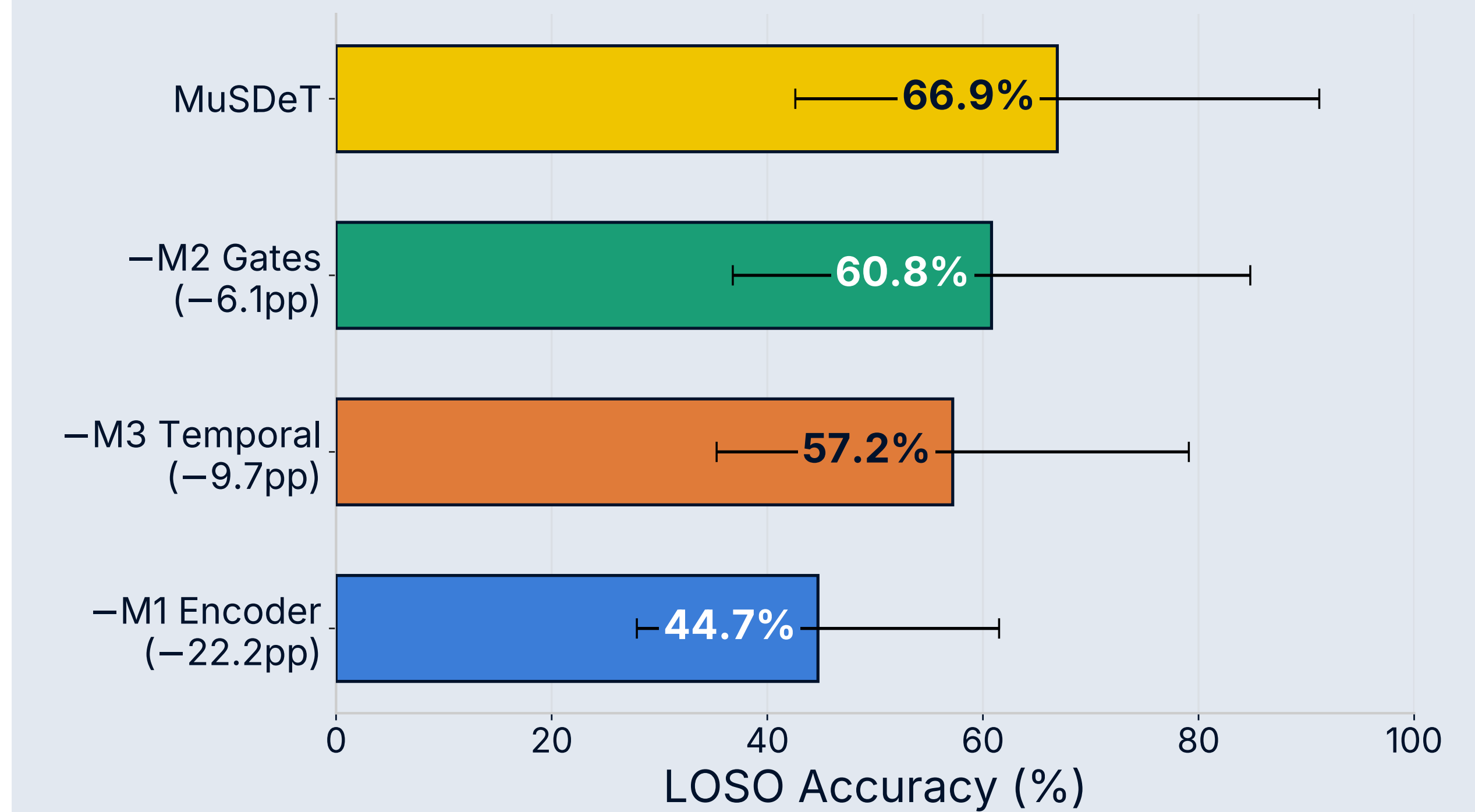
RESULTS

MuSDeT reaches 66.9% LOSO on WESAD — the worst seed stays above all baselines.

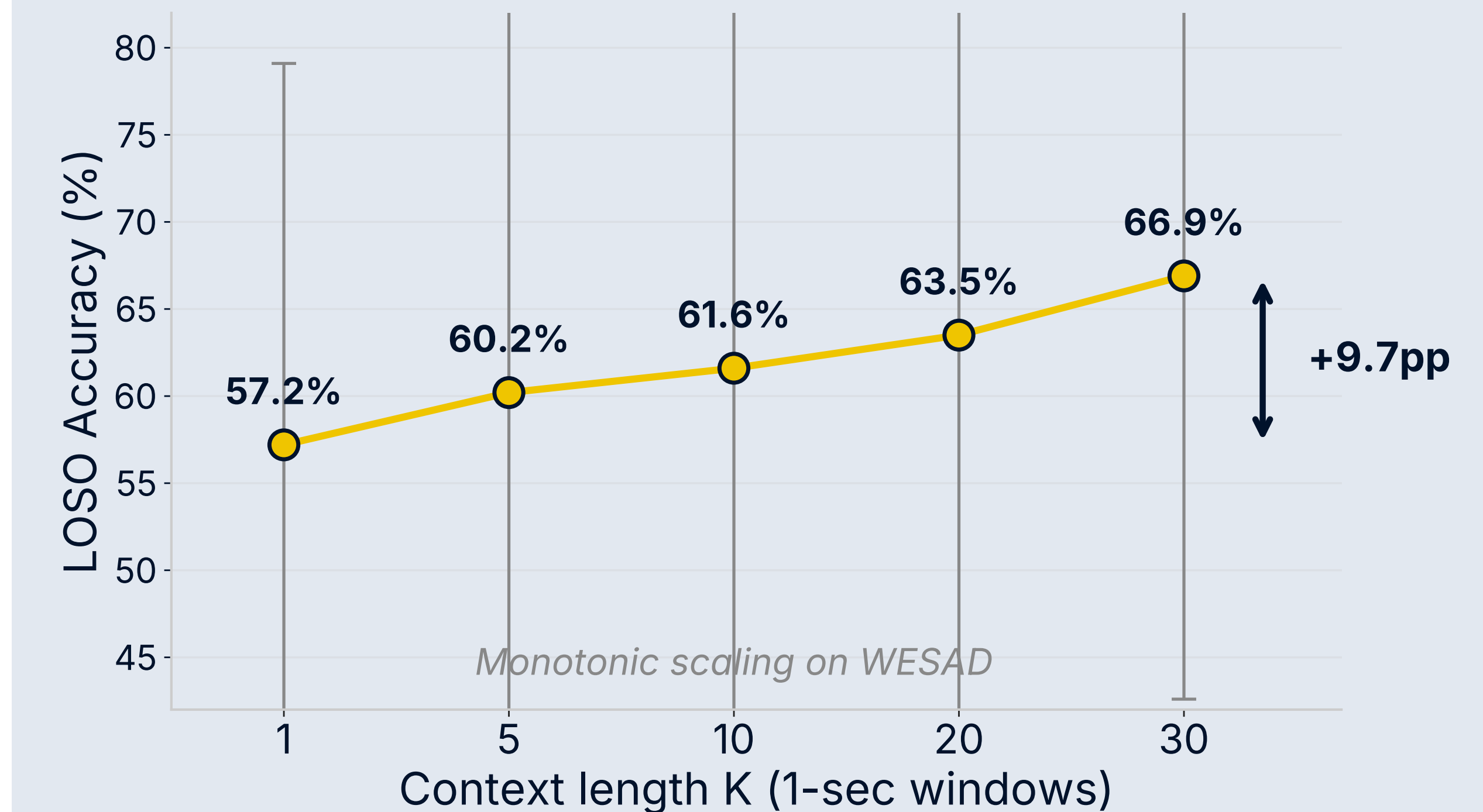
Protocol: leave-one-subject-out, zero/near-zero window overlap, 5 seeds.



ABLATION & SCALING



−M1 Encoder: single-kernel CNN −M2 Gates: equal weights −M3 Temporal: $K=1$.



Accuracy rises with temporal context length K , then plateaus.